If you want, I can create and give you a downloadable .docx MS Word file with this formatting.

YES

Random experiment:- If in each trial of an experiment conducted under identical conditions, the outcome is not unique, but may be any one of the possible outcomes, then such experiment is called a random experiment.

Example of Random experiment:-

Ex.1) Tossing a coin (H, T)

Ex. 2) Throwing a die (1,2,3,4,5,6)

Ex.3) Selecting a card from a pack. (Only one card, Like King, Queen, Jack, Ace, means Not Unique or same)

Ex.4) Bag Contain 3 red balls, 4 white balls,2 balls are drawn at random (2 red, 2 white, 1 red - 1 white) (Not Unique)

Outcomes:- the result of the random experiment is called outcomes.

Trial or Event :- Any particular performance of a random experiment is called trial and outcome or combinations of outcomes are termed as events.

Example :-

If a coin is tossed repeatedly, result is either Head or Tail, is called as Trial or Event

Exhaustive events :- The total number of possible outcomes of a random experiment.

Example 1) In tossing a coin, there are two exhaustive cases - Head, Tail. (Out comes H, T)

Example 2) In throwing a die, there are 6 exhaustive cases.(Out comes 1,2,3,4,5,6)

Example 3) Bag Contain 3 red balls, 4 white balls, 2 balls are drawn at random

(2 red, 2 white, 1 red - 1 white)

means Total ball = 5

You pick = 2 i.e. Exhaustive events = $^5C_2$

Mutually Exclusive events :- Two events are said to be mutually exclusive if the happening of any one of them exclude the happening of all the others.

Example: In throwing a die, all the 6 faces are mutually exclusive.

In tossing a coin, events Head and Tails are mutually exclusive.

Independent Events: Two events are said to be independent, if happening of one is not affected by the happening of others.

Examples: In tossing an unbiased coin, event of getting a head in first toss is independent of getting a head in the second, third and subsequent thrown.

Equally Likely event: Two events are said to be equally likely events if chances of occurrence of events are equal.

Mathematical (or Classical or A Priori) Probability

If a random experiment results "N" exhaustive (Total Number), exclusive (Disjoint) and equally likely outcomes then probability of event A is defined as

$$\mathbf{P(A) = \frac{Number\ of\ Favourable\ Cases\ Total}{Number\ of\ Exhaustic\ Cases}}$$

Basic Probability Formula :

$$\mathbf{P(A) = \frac{Number\ of\ Favourable\ Cases\ Total}{Number\ of\ Exhaustic\ Cases}}$$

Complement Rule :

$$\mathbf{P(A) = 1 - P(A')}$$

Probability – Two Unbiased Dice

Example 1] Two unbiased dice are thrown. Find the probability that:
(i) Both the dice show the same number
(ii) The first die shows 6
(iii) The total of the numbers on the dice is 8

Solution : Total exhaustive cases = 6 × 6 = 36

(i) Both the dice show the same number
Favorable cases: {(1,1), (2,2), (3,3), (4,4), (5,5), (6,6)}

Required probability = $\frac{6}{36} = \frac{1}{6}$

(ii) The first die shows 6
Favorable cases: {(6,1), (6,2), (6,3), (6,4), (6,5), (6,6)}

Required probability = $\frac{6}{36} = \frac{1}{6}$

(iii) The sum of the numbers on the dice is 8
Favorable cases: {(2,6), (6,2), (3,5), (5,3), (4,4)}

Required probability = $\frac{5}{36}$

Probability – Four Cards Drawn From a Pack of 52 Cards

Example 2] Four cards are drawn at random from a pack of 52 cards. Find the probability that:

(i) They are a king, a queen, a jack and an ace.

(ii) Two are kings and two are queens.

(iii) Two are black and two are red.

Solution : Total number of cards in a pack = 52

Number of cards drawn = 4

Total number of possible cases = $^{52}C_4$

(i) One king, one queen, one jack and one ace

Number of kings = 4

Number of queens = 4

Number of jacks = 4

Number of aces = 4

Favorable cases = $^{4}C_1 \times {}^{4}C_1 \times {}^{4}C_1 \times {}^{4}C_1$

$= 4 \times 4 \times 4 \times 4$

$= 256$

Required probability = $\frac{256}{^{52}C_4}$

(ii) Two kings and two queens

Favourable cases = $^{4}C_2 \times {}^{4}C_2$

Required probability = $(^{4}C_2 \times {}^{4}C_2) / {}^{52}C_4$

(iii) Two black cards and two red cards

Number of black cards = 26

Number of red cards = 26

Favourable cases = $^{26}C_2 \times {}^{26}C_2$

Required probability = $(^{26}C_2 \times {}^{26}C_2) / {}^{52}C_4$

Probability – Balls Drawn from an Urn

Example 3] An urn contains 6 white balls, 4 red balls and 3 black balls. If 3 balls are drawn at random, find the probability that:

(i) Two of the balls drawn are white.

(ii) None of the balls is red.

Solution : Number of white balls = 6

Number of red balls = 4

Number of black balls = 3

Total number of balls = 13

Number of balls drawn = 3

Total number of possible cases = $^{13}C_3$

(i) Two of the balls drawn are white

This can happen in the following ways: (and = Independent = +)

- 2 white and 1 red
- 2 white and 1 black

Favorable cases = $(^6C_2 \times {}^4C_1) + (^6C_2 \times {}^3C_1)$

Required probability = $[(^6C_2 \times {}^4C_1) + (^6C_2 \times {}^3C_1)] / {}^{13}C_3$

(ii) None of the balls is red

This means all balls are chosen from white and black balls only.

Number of white and black balls = 6 + 3 = 9

Favorable cases = $^9C_3$

Required probability = $^9C_3 / {}^{13}C_3$